

Qatar-4b

R-2717

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_260723 · created 2026-09-01T01:50:13+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461244.8859 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.0035
Transit depth (Rp/Rs)^2	3.38 +/- 0.13 [%]
Orbital Inclination (inc)	86.57 +/- 1.32
Airmass coefficient 1 (a1)	0.789 +/- 0.036
Airmass coefficient 2 (a2)	0.212 +/- 0.052
Scatter in the residuals of the lightcurve fit is	4.69 %
Best Comparison Star	#2 - [609, 230]
Optimal Method	PSF photometry
Transit Duration (day)	0.0844 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.0035 vs 0.138 (deviation 6.08σ)
Duration fit vs published	0.0844 d vs 0.0892 d (deviation 0.49σ)
Mid-time O-C	-9.5 min
Depth SNR	24.3
Residual scatter	4.69 %

Light curve

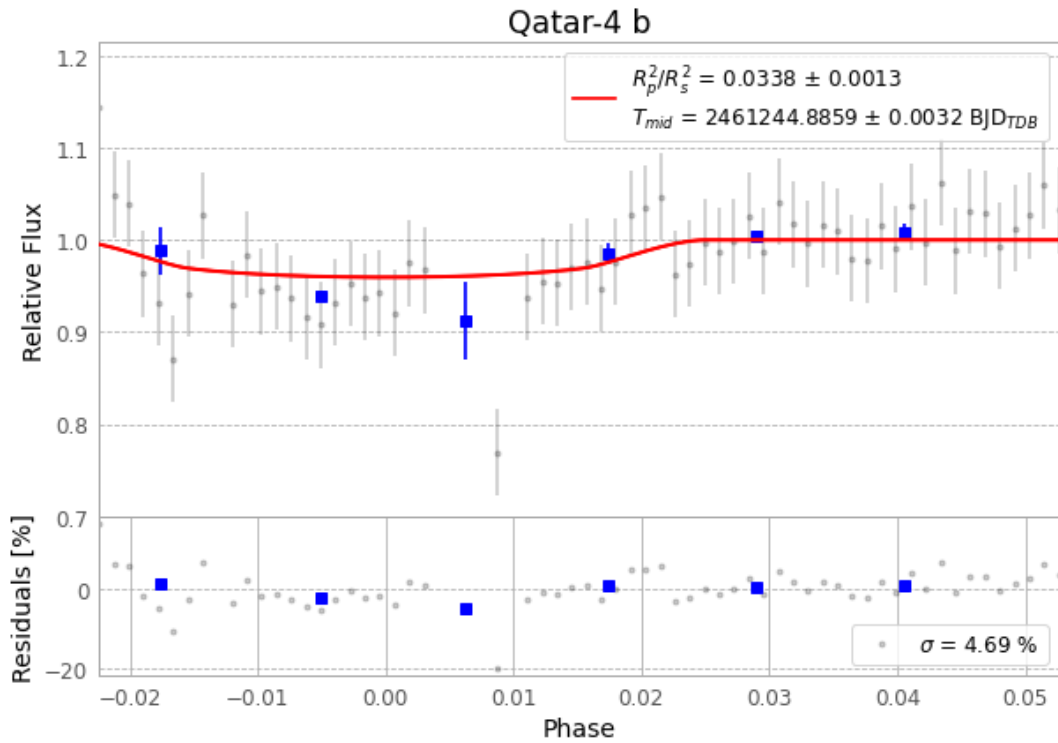


Figure 1 — FinalLightCurve_Qatar-4 b_2026-07-23.png (SHA-256 e64074db761b8236...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [222, 282], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 13e045573882d1c2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), Qatar-4260723081516.FITS → Qatar-4260723113015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-260723.log (fddc9a3eccf2...), FinalLightCurve_Qatar-4 b_2026-07-23.png (e64074db761b...), FinalLightCurve_Qatar-4 b_2026-07-23.csv (8d8c4de94dc8...)

Compute resources

Wall time 123.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:50Z.